deliberations and decisions, to be used later for discovery and in some cases for litigation. The significant issue with a smaller membership and consequently with private blockchain is the 51 percent attack. In this case, it is potentially easier for a few members to collude and make the chain inconsistent.

All three types of public blockchains--permissionless, permissioned, and private--are relevant in the blockchain application domains. They differ mainly in the way membership is determined. With closed membership, it is possible to implement a more efficient consensus mechanism than the POW of Bitcoin, which consumes an enormous amount of power. But in a closed, private system, you are reverting to trusting a few designated participants, as in a centralized system. You must weigh these facts when deciding which type of blockchain is appropriate for a given problem. Whether the blockchain is public, private, or permissioned, you need security mechanisms to protect the data.

12.8 Confidentiality

Many people equate privacy with confidentiality. Confidentiality is different from privacy; it is about keeping details (or metainformation) about the transaction closed. In some situations, transactions have to be confidential. Consider a patient-doctor relationship. If there were ten transactions between a patient and healthcare provider in a day, this information might convey something, even if the content of the transactions is kept private. The fact that ten transactions occurred between the parties implies something.

Let’s explore how open the transactions are in the Ropsten network you used to deploy your Dapps.